Protogen

Protogen is a Streamlit application for automating liquid handling protocols for Opentrons OT-2 and Janus robotic systems.
It generates protocols for DNA assembly and allows users to intuitively define components, volumes, and well positions.

Features

- **Excel-based source plate loading**
- **Effimodular experimental design**
- **Automatic OT-2 and Janus protocol/mapping generation**
- **Required volume and plate feedback**

Installation

Protogen is written in Python 3.9.

```
pip install streamlit pandas opentrons openpyxl
```

Usage

Run the following command:

```
streamlit run protogen.py
```

1. File Upload (Input File Format)

Protogen uses Excel files (`.xlsx`) as input. Each sheet contains a table in a 96-well plate format, with the name of the corresponding component recorded in each well.

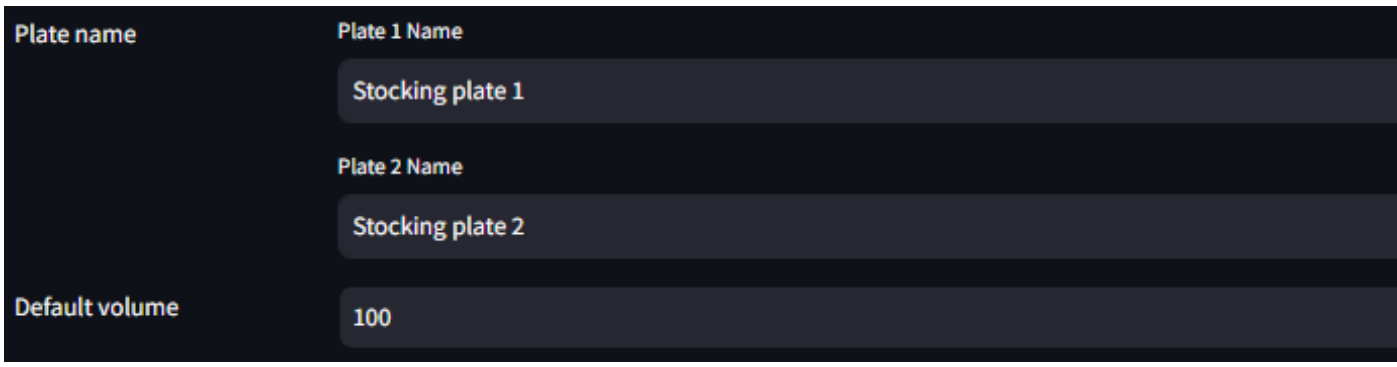
	1	2	3	…	12
A	(P)1	(C)1	(T)1		
B	(P)2	(C)2	(T)2		
C	(P)3	(C)3	(T)3		
…					
H	(P)8	(C)8	(T)8		

- Each well (e.g., A1, B1, C1) contains the names of TU Parts such as Promoter, CDS, and Terminator.
- Unused wells should be left empty.

- Each Part is categorized using a header (P, C, T, N) at the beginning of its name, allowing the system to recognize Part positions automatically.
- Parts with identical names are treated as the same Part.

Header	Part
(P)	Promoter
(C)	CDS
(T)	Terminator
(N)	Connector

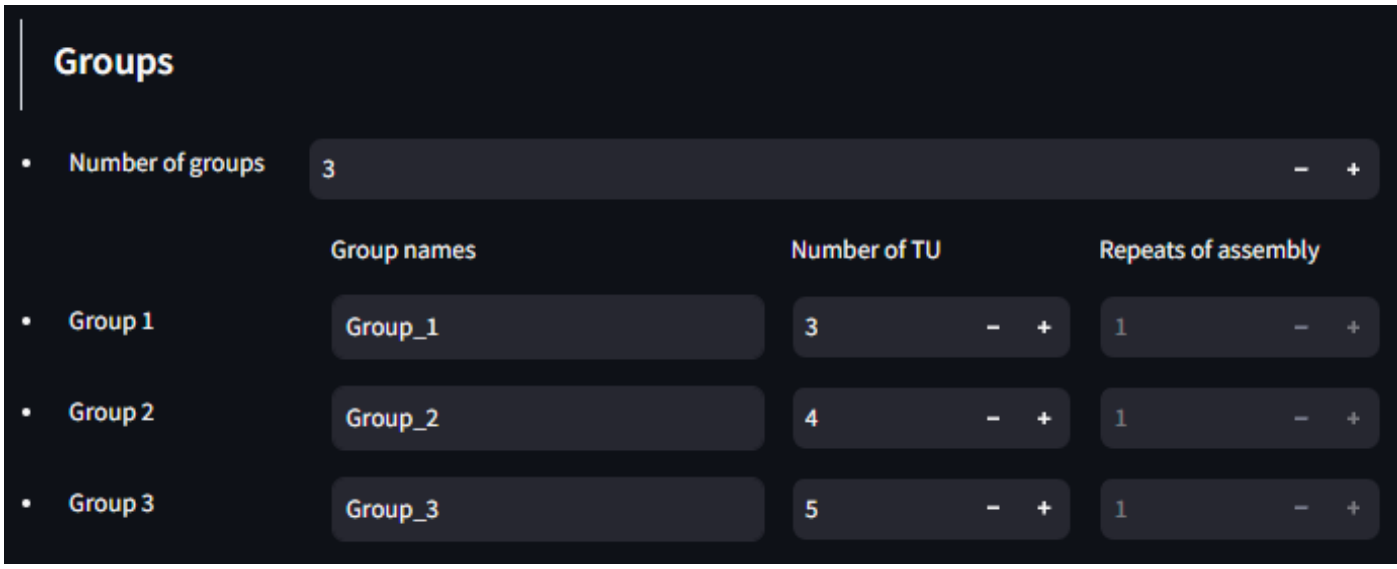
2. Plate and Volume Settings



- Assign a name to each plate from the uploaded file.
- Set a common volume for wells without volume data.

3. Lv1 Design

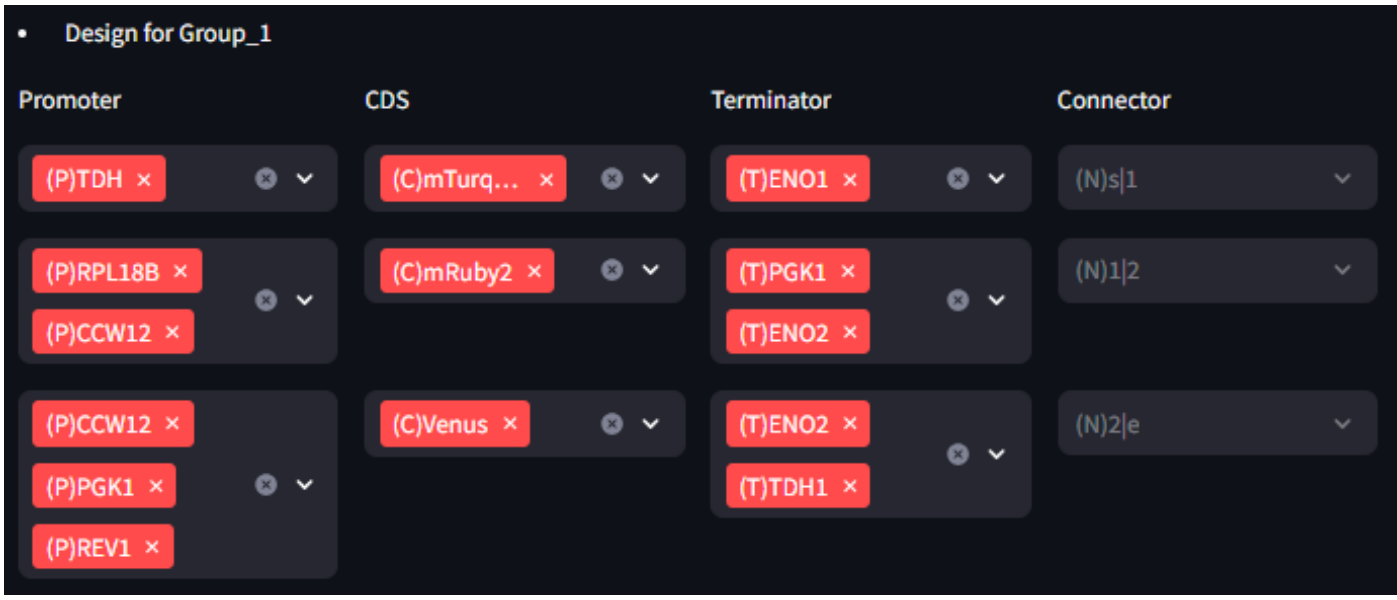
Groups



This function enables multi-plasmid design. Each group is designed independently.

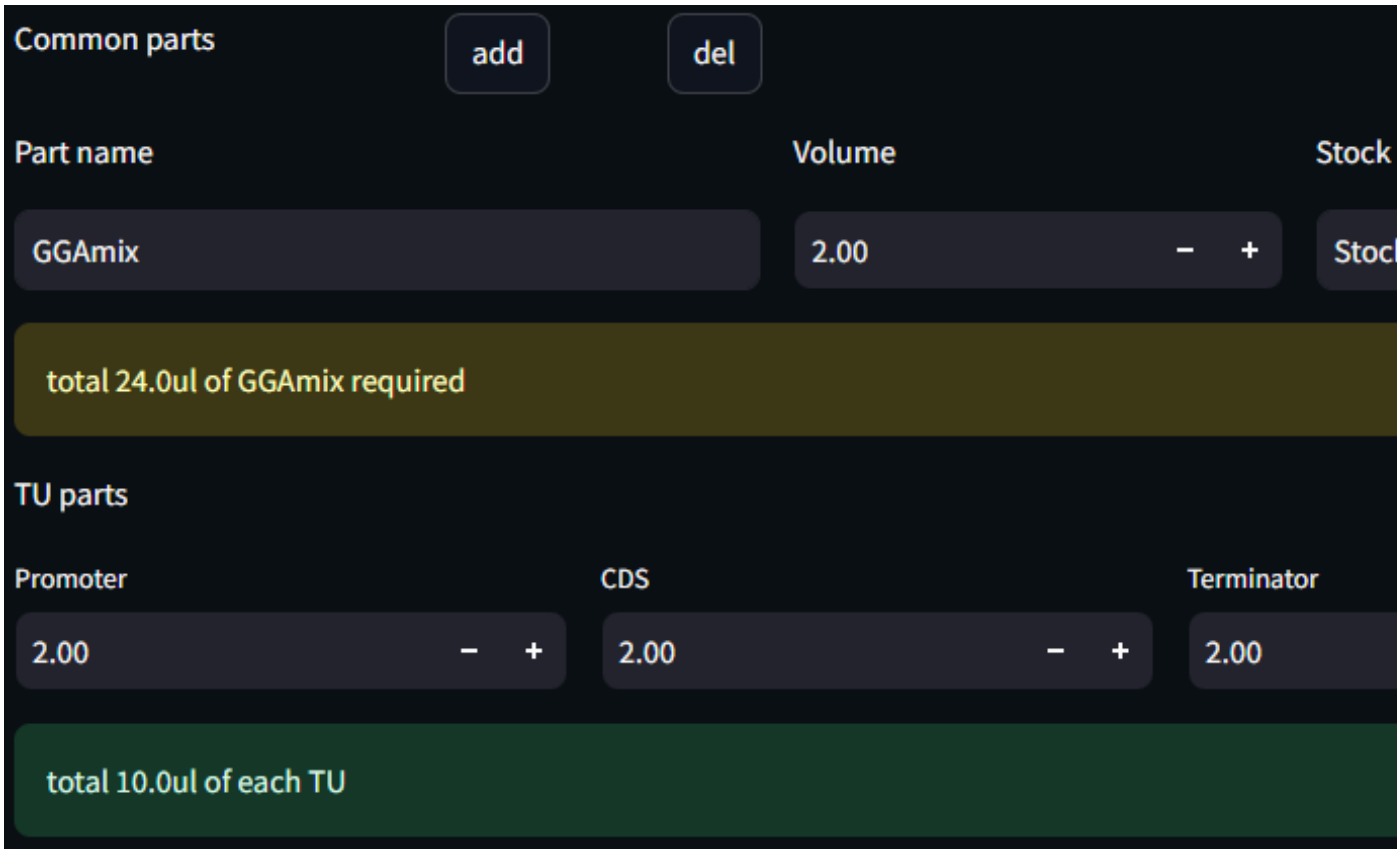
- **Number of groups**: Defines the number of groups dividing the Transcription Unit (TU).
- **Group names**: Assigns names to each group (duplicates may cause errors).
- **Number of TU**: Specifies the number of TU each group contains.

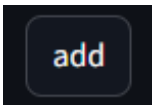
TU Design



- Defines detailed TU specifications for each group.
- Each TU includes one CDS, and if multiple Promoters or Terminators are specified, all possible combinations are calculated to proceed with Lv2 Design.
- Overlapping Part combinations are excluded.
- Connectors are fixed.

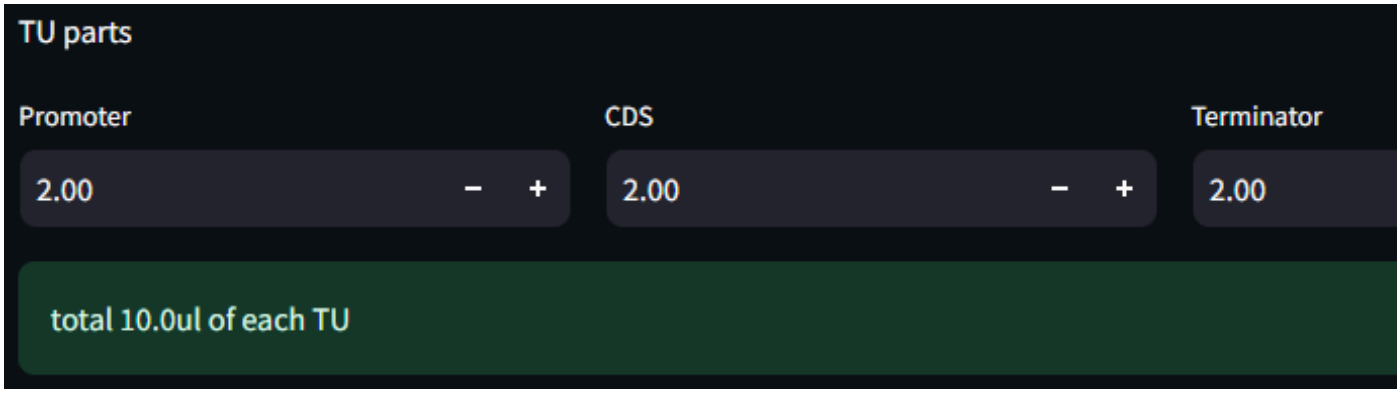
Volume Setting



- **Common parts**: Defines Parts to be included in all TU.
- Use 

and

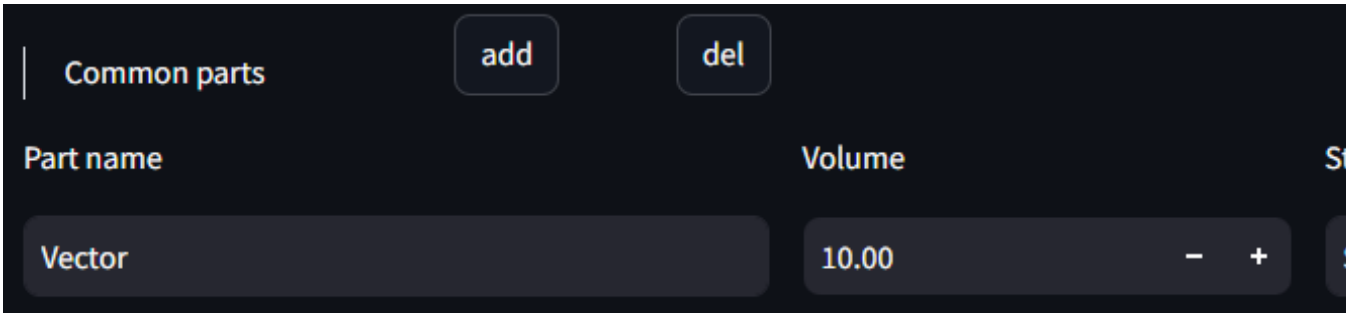
to add or remove Parts.
- Part names can be freely assigned, and the specified stock plate and stock location are used to load the corresponding Part.
- **If the Part name overlaps with an already loaded stock plate, conflicts may occur.**
- Entering the Name and Volume calculates the total required amount based on the number of Lv1 designs.

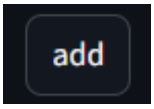


- **TU Parts**: Sets the volume for each Part (unit: μL).
- The total volume for each TU is calculated after setting.

4. Lv2 Assembly Design

- **Lv2 Volume Setting**: Defines the volume per TU and the dead volume for Lv2 assembly.
  - Each combination includes the defined volume per TU.



- **Common Parts**: Defines Parts to be included in all TU.
  - Use 

and

to add or remove Parts.
  - Part names can be freely assigned, and the specified stock plate and stock location are used to load the corresponding Part.
  - **If the Part name overlaps with an already loaded stock plate, conflicts may occur.**
  - Entering the Name and Volume calculates the total required amount based on the number of Lv2 designs.

Calculation Method

Each TU contains one CDS. If multiple Promoters or Terminators are specified, all possible combinations are calculated to generate the Lv2 Design.

- In the given example, 1+2*2+3*2 = 11 TU designs are created.

	name	plate	well	volume	note	0
0	Group_1_1	Dest_02_1	A1	34	Group_1_1	(P)TDH-(C)mTurquiose2-(T)ENO1
1	Group_1_2	Dest_02_1	A2	34	Group_1_2	(P)TDH-(C)mTurquiose2-(T)ENO1
2	Group_1_3	Dest_02_1	A3	34	Group_1_3	(P)TDH-(C)mTurquiose2-(T)ENO1
3	Group_1_4	Dest_02_1	A4	34	Group_1_4	(P)TDH-(C)mTurquiose2-(T)ENO1
4	Group_1_5	Dest_02_1	A5	34	Group_1_5	(P)TDH-(C)mTurquiose2-(T)ENO1
5	Group_1_6	Dest_02_1	A6	34	Group_1_6	(P)TDH-(C)mTurquiose2-(T)ENO1
6	Group_1_7	Dest_02_1	A7	34	Group_1_7	(P)TDH-(C)mTurquiose2-(T)PGK1
7	Group_1_8	Dest_02_1	A8	34	Group_1_8	(P)TDH-(C)mTurquiose2-(T)PGK1
8	Group_1_9	Dest_02_1	A9	34	Group_1_9	(P)TDH-(C)mTurquiose2-(T)ENO2
9	Group_1_10	Dest_02_1	A10	34	Group_1_10	(P)TDH-(C)mTurquiose2-(T)ENO2
10	Group_1_11	Dest_02_1	A11	34	Group_1_11	(P)RPL18B-(C)mTurquiose2-(T)ENO1
11	Group_1_12	Dest_02_1	A12	34	Group_1_12	(P)RPL18B-(C)mTurquiose2-(T)ENO1
12	Group_1_13	Dest_02_1	B1	34	Group_1_13	(P)RPL18B-(C)mTurquiose2-(T)ENO1
13	Group_1_14	Dest_02_1	B2	34	Group_1_14	(P)RPL18B-(C)mTurquiose2-(T)ENO1
14	Group_1_15	Dest_02_1	B3	34	Group_1_15	(P)RPL18B-(C)mTurquiose2-(T)ENO1
15	Group_1_16	Dest_02_1	B4	34	Group_1_16	(P)RPL18B-(C)mTurquiose2-(T)ENO1
16	Group_1_17	Dest_02_1	B5	34	Group_1_17	(P)RPL18B-(C)mTurquiose2-(T)PGK1
17	Group_1_18	Dest_02_1	B6	34	Group_1_18	(P)RPL18B-(C)mTurquiose2-(T)PGK1
18	Group_1_19	Dest_02_1	B7	34	Group_1_19	(P)RPL18B-(C)mTurquiose2-(T)ENO2
19	Group_1_20	Dest_02_1	B8	34	Group_1_20	(P)RPL18B-(C)mTurquiose2-(T)ENO2

5. Outputs

- **Select OT-2 or Janus**: Choose the robotic system to execute.
- **Automatic protocol generation**: Generates OT-2 or Janus protocols based on selected settings.
- **Check outputs**: Review final output well positions and used reagents.
  - **OT-2 Protocol**: Python script formatted for execution on Opentrons.
  - **Janus Protocol**: CSV mapping file for Hamilton Janus.
  - **Updated inventory data**: Dataframe with updated volume calculations.